Overall Course Goal

Build software

Overall Course Goal

Build software

With other people

Overall Course Goal

Build software

With other people

Effectively

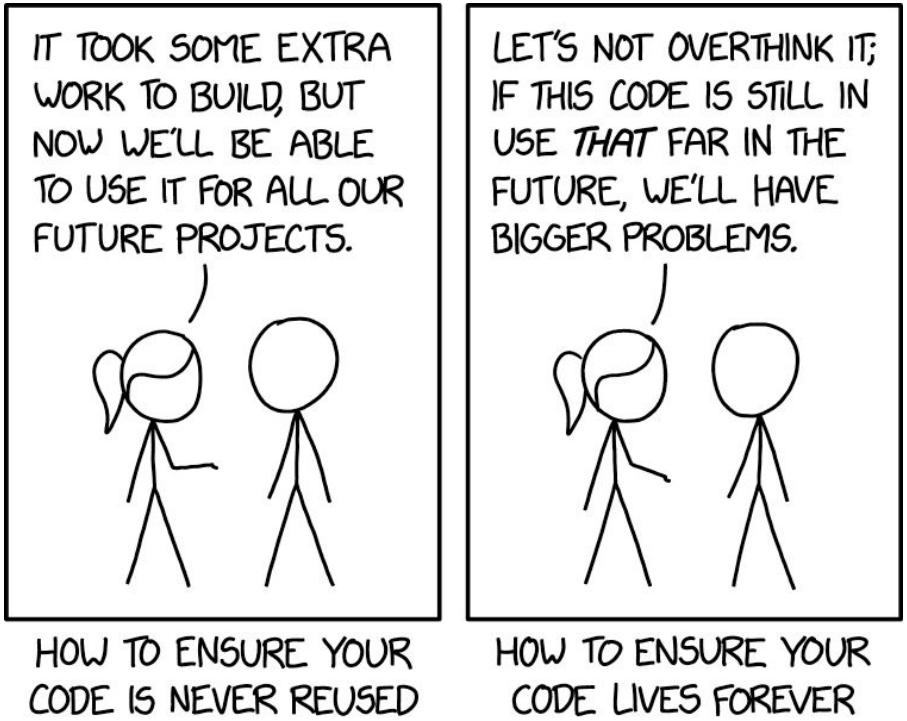
Overall Course Goal

Build software

With other people

Effectively

Over time



Course Philosophy

Learn the Theory (Version Control, Dependency Management, Design Patterns, etc)

- Build a useful foundation

Practice with Specific Tools (Git, Gradle, Java, etc)

- Current industry standard versions because why not

Development Ecosystem

Version Control + Local Checkout + Build System Management + IDE = Success

Specifics:

Github + Github Desktop + Gradle + Eclipse

Supported vs. Allowed

Common distinction in industry

Allowed vs Not Allowed: Security, Dogfooding

Supported vs Not Supported: Standard workflow, existing expertise

Are Those Exact Specifics Required?

Github = Required (others not allowed)

Github Desktop = recommended, but any Git client will work

- Supported: Github Desktop, command-line git
- Can use multiple clients at the same time, low risk to experiment here

Gradle = Required

- Maven: technically allowed, but a lot more work, **unsupported**

Eclipse = **strongly** recommended

- VSCode: not allowed, because it's missing required features
- IntelliJ: allowed, **unsupported** (FYI, they embrace 'move fast and break things')

Basic Workflow

Your local copy
on your laptop

Changes
to the files

VCS
(Github)

Basic Workflow

Git Client
(Github Desktop)
syncs to cloud

Basic Workflow

All repository files are
stored in the File System

Basic Workflow

Dependency Management (Gradle) reads the files and creates **local metadata files**

This includes IDE-specific configuration (either Eclipse or IntelliJ)

Basic Workflow

Your IDE (Eclipse) can be configured locally without affecting anyone else

Demo! (Time Permitting)

Grab the starter code

Import into IDE

Commit changes

Invite users to a repo

Code Reviews: Cheap Way to Avoid Bugs

Software equivalent of editing a paper

Rough estimate:

A bug that takes 1 minute to fix if flagged in a code review will take:

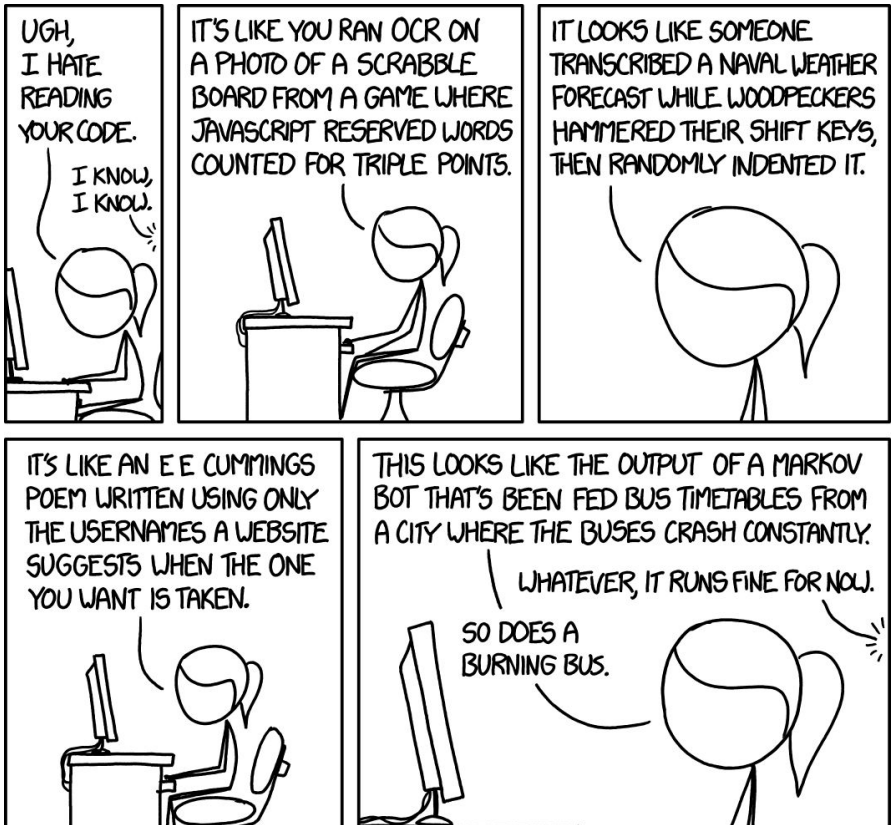
- 10 minutes to fix if flagged in a manual testing pass
- 100 minutes to fix if shipped to production

Code Reviews: Cheap Way to Avoid Bugs

…But Only If Done Right

Best practices make Code Reviews

- Lightweight
- Effective at spotting issues
- Pleasant



Structuring Code For Code Review

Style **conventions** are aspects of code not enforced by the compiler, but which make code much more approachable:

Ex:

```
int getWidth() {
    return width;
}
```

and

```
rectangle.getWidth()
```

vs

```
 int Width(){return a;}
```

and

```
b.Width()
```

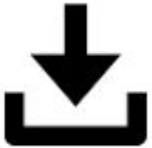
Structuring Code For Code Review

But wait, those two bits of code looked equally easy to read!

Conventions develop value **over time**

Compare two icons:

vs



The first time you encounter a 'Save' icon, either of those makes as much/little sense as the other.

Once you've seen 50, the one on the left is much easier to recognize

Structuring Code For Code Review

Some typical conventions (not exhaustive):

- methods and variables begin with a lowercase letter (int width, void doThing())
- constants are all caps (int x = DEFAULT_VAL;)
- class names begin with a capital letter (public class Math {})

Some optional conventions: pick something and be consistent in the code base

- tabs vs 2 spaces vs 4 spaces
- { on a newline vs with a space

Structuring Code For Code Review

Beyond surface-level conventions, make sure to:

Include comments on **why** some code exists, not what it does:

```
// adds 2 to x
x += 2;
```

vs

```
// We're only considering even numbers
x += 2;
```

Structuring Code For Code Review

Remember good **object oriented** practices

A class should have **one** data concept, a method should have **one** task/behavior; if you can't describe it with a single sentence, or if you're using the word "and", be suspicious (this is a **code smell**)

Ex: getWidthAndHeight() vs getWidth(), getHeight();

Structuring Code For Code Review

Not a hard-and-fast rule:

```
 // Represents a deck of cards. Supports shuffling and dealing.
public class Deck {
   List<Card> cards;

   public void shuffle() {}

   public void deal() {}
}
```

There are multiple sentences and an 'and', but there's a single 'data' concept and the two 'behaviors' are both closely related to the data and in separate methods

Structuring Code For Code Review

```
// Represents card numbers and suits
public class Deck {
   List<Integer> cardNumbers;
   List<Integer> cardSuits;
}
```

The 'and' here is in fact a problem - we're not properly **encapsulating** the data. The two items being 'and'ed together are closely related, which for data means they should be grouped in their own class (Card)

```
/* Represents a deck of cards. Supports shuffling and coordinates placing pieces on a game board
*/
public class Deck {
   List<Card> cards;

   public void shuffle() {}

   public void initializeBoard() {}
}
```

Here, the 'and' is a problem because the two behaviors are very different; a game board has nothing to do with a deck of cards.

Structuring Code For Code Review

Code shouldn't be surprising:

Main.java:

```
System.out.println("Rectangle width: " +
                   rectangle.getWidth());
```

Rectangle.java:

```
public int getWidth() {
   setHeight(3); // ← eek!
   return width;
}
```

Structuring Commits For Code Review

- Manageable size (less than 200 lines of non-auto-generated code)
- Detailed description, descriptive summary message
- Single feature/bug/task

Giving a Code Review

Step 1: Grok the code

- Read the commit summary & description
- Skim through the whole change
- Do a second pass for deeper understanding
    - Make notes of anything that looks unclear
    - If necessary, sketch out a diagram/outline of what the change is doing

Giving a Code Review

Step 2: Comment on larger issues

Be nice! Criticism always feels harsher on the receiving end than the giving end

A code review is a **conversation** - for larger changes, expect to have multiple rounds of commits/reviews

Topics:

- Bugs: if something looks functionally broken
- Lack of testing: make sure there are at least basic tests
- Unclear code: missing comments, code that isn't self-documenting
- Structural issues (layers of abstraction/API design)

Giving a Code Review

Step 3: Proofreading for stylistic issues

For small issues: by convention, start with "**nit:**": Ex: "nit: add a newline between imports and class declaration"

- Whitespace/formatting
- typos in variable names
- suboptimal return statements:
  - Ex: "nit: return hasProperty; vs return hasProperty == true;"

These shouldn't block approvals, but should be fixed up before merging

Avoid **micromanaging**

- Determine if another way is genuinely better, or just different

Demo!

Leave inline comments

Resolve comments

Using a Linter

Code reviews that spend a lot of time on nits are sub-optimal

Fortunately, some formatting issues are easy to spot programmatically

A **linter** is something that cleans up code "lint" - the small bits of easily removed fluff to leave clean and polished code behind

checkstyle is an open source, easily configured version, which we'll be using.

Using Checkstyle With Github

We're going to hook up a custom checkstyle config to the status checks in Github

Every Pull Request will automatically trigger these checks

- Don't request a review/don't leave a review until these checks pass
- They should be very quick

This is an example of Continuous Integration (CI)

Note: there are lots of ways to call checkstyle, including from gradle and as an Eclipse plugin. Feel free to add additional calls/checks for your project!

Continuous Integration

Shorten the loop between code being written and being deployed to production

Automates error detection, shipping, and rollbacks

Complexity ranges from very simple (checkstyle checks) to very complex (automated canary rollouts with rollbacks if certain metrics spike)

Benefits:

- Faster feedback
- Easier to launch features
- Higher quality code

Pair Programming

Combines the coding and code review steps

Requires 2 people looking at the same screen with low-friction instant communication (ideally in-person, but screenshare with discord also works; email does **not** count as low-friction instant communication, though)

One person types, the other reviews

Good for the first several commits of a large, complex architecture

- avoids latency and context switching from code reviews

When committing, note in the **commit description** that the code was pair-programmed (and with who); this should line up with the person who then **approves** the pull request

Where/How to Write Code

Ex: Writing a 20 page research paper

You **could** use a plain text editor, but you probably wouldn't

Instead, google docs/libre office/ms word:

- Spell check
- Grammar check
- Formatting
- Text suggestions

These tools are **integrated** into your word processing **environment**

Where/How to Write Code

Same for code! Use an **I**ntegrated **D**evelopment **E**nvironment (IDE)

- Constant compilation/compile error readout
- Warnings
- Syntax highlighting
- Auto-formatting
- Auto-completion/auto-generated code

IDE for Java

Recommended: Eclipse

- Fully featured
- Easy to migrate from this to other tools

Optional: IntelliJ

- Industry standard
- Not so great if you don't have a build tools team

Do Not Use: Visual Studio Code (VSCode)

- Buggy
- Slow
- Poor Gradle integration

Unlock the Power of the IDE

Emacs/notepad is fine for C code, but modern languages are designed to be written with an IDE

- Solves the complexity to type vs complexity to read tradeoff
- Minimizes "toil"
- Allows for easy navigation of object-oriented code

Shortcuts in the IDE are HUGE for productivity, they are absolutely worth learning

Deterministic vs Stochastic Code Generation

Deterministic: Compiler, Many IDE tools

- No need to double-check output (“Just glance through the .class bytecode to make sure it’s right” said no one ever)
- Solves specific problems

Non-deterministic (stochastic): LLMs, other ML models, stack overflow

- Must double-check output
- Can attempt any problem, but no guarantee of successfully solving

⇒ For any problem that **can** be solved deterministically, **prefer that option**

This requires **knowing when those options exist**

Organize Imports

Eclipse shortcut: ctrl + shift + o

Doesn't just "organize" existing imports

Searches the classpath for any unreferenced types, and attempts to add them

Classic API design:

- Import package structure is an **implementation detail**
- The library classes/interfaces are the **API**

Organize imports = "Hey IDE, I want this class. Go find it for me"

Auto-Refactor

Eclipse shortcut:

Rename: alt + shift + r

Extract method: alt + shift + m

Rename Classes/parameters/methods/interfaces

Extract methods

⇒ Reduces the cost of readability/maintainability fixes

Autocomplete

Eclipse shortcut: ctrl + space

Long variables/methods

⇒ Prioritize readability without a hit to productivity

Forgot a method? The IDE is here to help!

Eclipse shortcut: type . and wait

⇒ Efficiently use a large number of libraries

Navigate Object Hierarchy

Eclipse shortcut: ctrl + click

Object Oriented Programming has tons of classes, interfaces, and instance variables that all interact across files

Navigating in the file structure is slooooooow

⇒ context switching is expensive

⇒ reasoning about the whole system is difficult

Better: click-through navigation in the IDE

And Many More!

Deterministic generation (no generative AI, please!):

- hashcode/equals/toString
- getters/setters
- try/catch blocks
- unimplemented methods

Auto-format files

"On save" actions

Etc.

Let's Review!

What you should know from this week that might be on an exam:

Version Control: What is it, what are branches, when would you use a feature vs. release branch

Github Workflow: What is a Pull Request, what is branch protection, what are status checks

Code Reviews: Why have them, the 3 features of a good code review process, tips for structuring code and commits for review, and the 3 steps in giving a code review

Style Conventions: What are some common ones, why are they important

Continuous Integration: What are some examples, what are some benefits

Pair Programming: What is it, and what problem with standard code reviews does it solve

Build Tools: Common tools, why they're useful

IDEs: Common IDEs, why they're useful